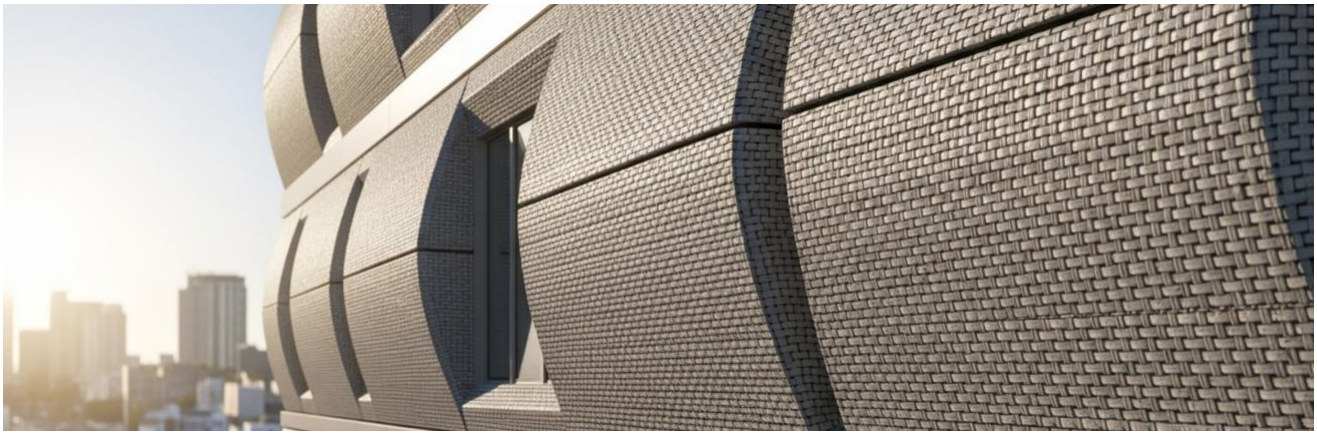


The "Aero-Weave" Facade System

The Common Problem: Urban Heat Island (UHI) Effect

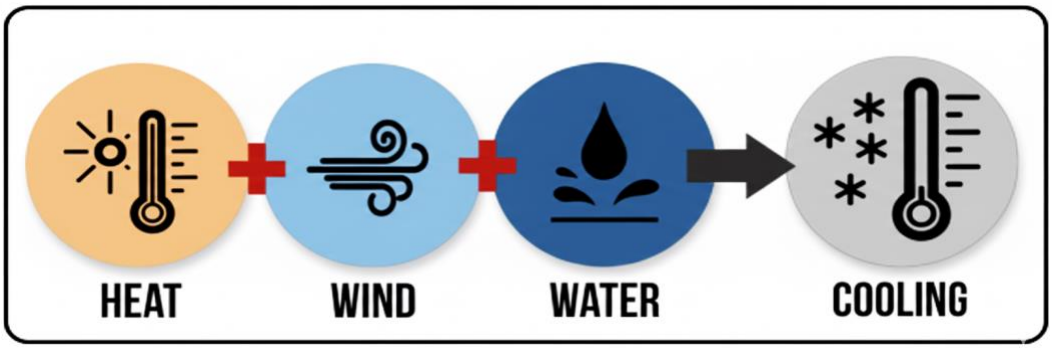
All modern cities suffer heavily from the Urban Heat Island (UHI) effect. Concrete flats absorb massive amounts of heat during the day and radiate it at night. This heat absorption and retention create a reliance on air conditioning, which consumes huge amounts of energy and dumps back heat and toxins into the environment, creating a vicious cycle of heating and energy consumption.

Factors affecting the problem

Factor	Effect
Concrete	Massively absorb and store heat
AC Reliance	High energy consumption
AC Exhaust	Dumps hot air back into the environment

The Innovative Solution: The "Aero-Weave"

The "Aero-Weave" is a biomimicry-inspired, retrofittable facade designed to be attached to the exterior of existing apartments and buildings, specifically targeting high wind and sun facing sides.
Instead of just blocking light like standard louvers, it uses thermodynamics and fluid dynamics applying the concept of Bernoulli's principle and its subsequent Venturi effect to passively cool buildings and reduce dependency on AC systems. The basic principle is to use water as an effective coolant.



The Overview of the Solution

The architectural landscape is heavily reliant on materials that exacerbate this issue. This project delves into a solution that makes use of basic geometry, shapes, and physics, employs material science and taps into the potential of water as the environment-friendly coolant.

1. The Venturi Effect

The facade primarily features funnel-shaped panels.

- **Mechanism:** As strong, hot and humid winds hit the Aero-Weave facade, the air is forced through these narrower channels that are cooled through pipes running cold water. Cooled air through the heat exchange process is then directed into the interior of the apartment without the use of pumps or fans.
- **Result:** This acceleration of airflow due to the Venturi Effect, creates a cooling breeze that is directed into the corridors and apartments situated behind the facade.
- **Principle:** Using Bernoulli's principle and the Venturi effect, accelerating air flow and eliminating the need for fans to force air through the cooling section.

2. Core Inspirations

The conceptualization of Aero-Weave bridges material science and sustainable design:

- **Terracotta Evaporative Cooling:** The use of water-retaining ceramic tubes like the ancient practice of using porous earthenware for space cooling is successfully modernized by ventures like CoolAnt (https://coolant.co/).
- **Material Science:** The hydrocooling system integration is inspired by contemporary research into hydroceramics, which utilize clay and hydrogels as inexpensive passive cooling systems and evaporative facade materials developed for hot and humid environments.

3. Usage of sustainable sources for coolants

The use water as a hydro coolant to prevent any use of toxic materials or large amounts of electricity:

- **Rainwater Harvesting Integration:** The primary coolant running through the evaporative hydroceramic tubes is locally captured rainwater. By channelling rainfall directly into the facade's tube network, the system replaces the need for chemically synthesized and ozone-depleting refrigerants (like HFCs).
- **Greywater and AC Condensate Recycling:** To ensure continuous passive cooling even during dry spells, the system is designed to operate on a closed loop water cycle and absorb condensate back to the water resource. It can optionally use treated greywater to have net zero water loss. The facade features a capillary system which makes use of the AC condensate produced, which on average is 7.4 litres per house with a use of 8 hours (https://www.iosrjournals.org/iosr-jmce/papers/vol18-issue6/Ser-1/D1806012530.pdf) according to a study. This makes the system highly efficient keeping it saturated with enough coolant all year round.

4. Modular Retrofitting (Practical Architecture)

This design showcases a practical architecture. The proposal is not to tear down existing buildings but to improve their thermal efficiency and dependency on existing cooling systems:

- **3D Printed Terracotta panels:** Inspired by construction works of innovators like Tvasta (https://www.tvasta.construction/projects/), who use large scale 3D prints, the Aero-Weave modules are designed to be 3D printed using a specialized terracotta polymer blend. This approach helps make the panels in a short duration, utilizing terracotta's porosity for evaporative hydro cooling.
- **Installation:** It can be easily attached to the exterior of the buildings without needing to destroy the original standing structure.

5. Alignment with Sustainable Development Goals (SDGs)

Aero-Weave directly addresses several critical United Nations SDGs, demonstrating a commitment to global sustainability frameworks:

- **SDG 11 (Sustainable Cities and Communities):** By transforming existing HDB blocks into self-cooling structures, the project enhances the sustainability and thermal comfort of dense urban housing.
- **SDG 13 (Climate Action):** The system breaks the vicious cycle of air conditioning reliance, which actively reduces urban energy consumption and the corresponding carbon footprint.
- **SDG 12 (Responsible Consumption and Production):** Utilizing a recycled-polymer modular system ensures that the manufacturing process itself minimizes waste and promotes circular economy principles.

Design	Purpose
Passive Cooling	Reduces reliance on electrical air conditioning
Recycled Polymer and use of terracotta	Sustainable material choice
Retrofittable	Allows for easy upgrading of existing infrastructure

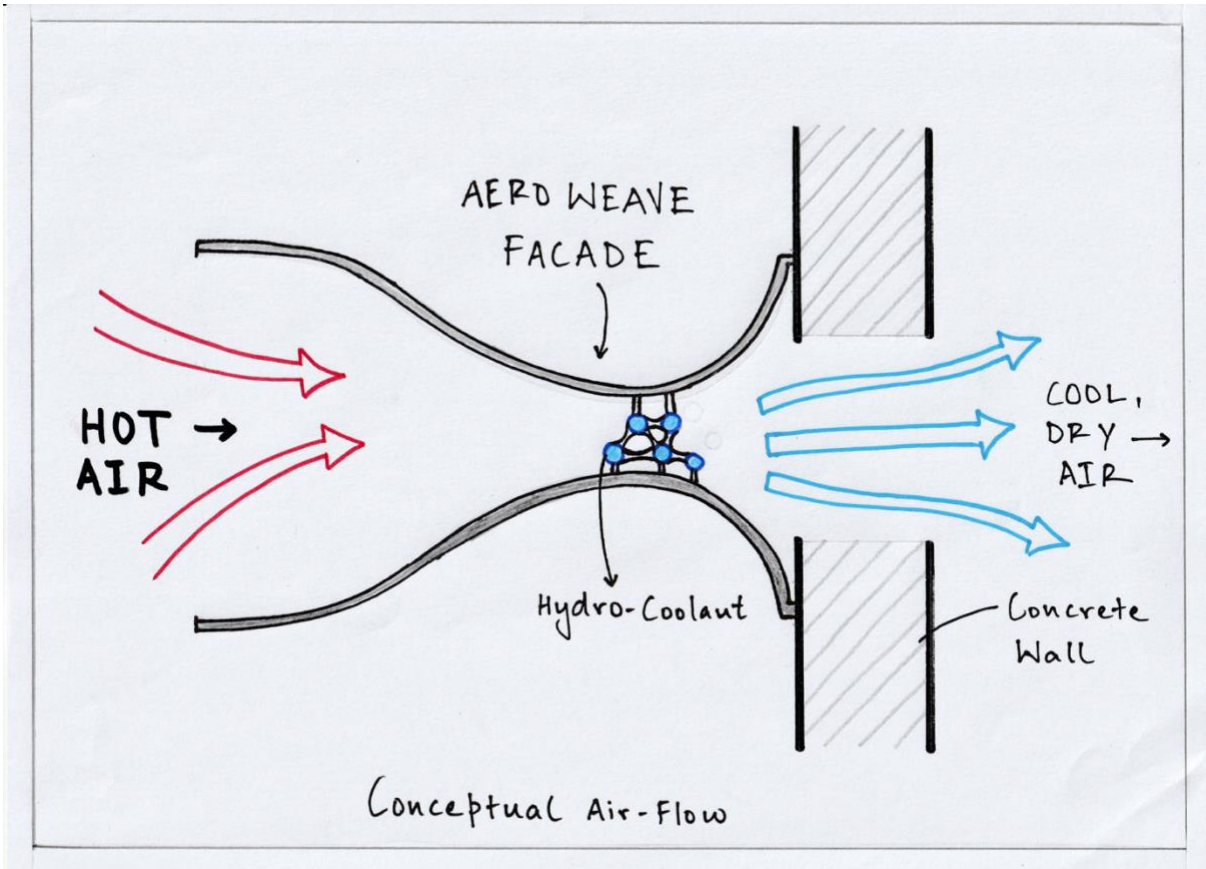
Motivation to start this project

The "Aero-Weave" project originated from my experience of living in Singapore during the 2019-2020 pandemic. Living there for nearly a year with my family confined to indoors 24/7 due to circuit breaker, I keenly felt the difficulty of staying comfortable without constant air conditioning. This period led to a significant increase in our AC usage.

While I didn't reflect on it much at the time, my recent pursuit of architecture has brought Singapore's growing dependence on air conditioning into focus. Motivated by this issue, and particularly after reading the article found here (https://edition.cnn.com/2023/06/09/asia/air-conditioning-singapore-climate-change-intl-hnk-dst), I decided to initiate this project.

The Idea

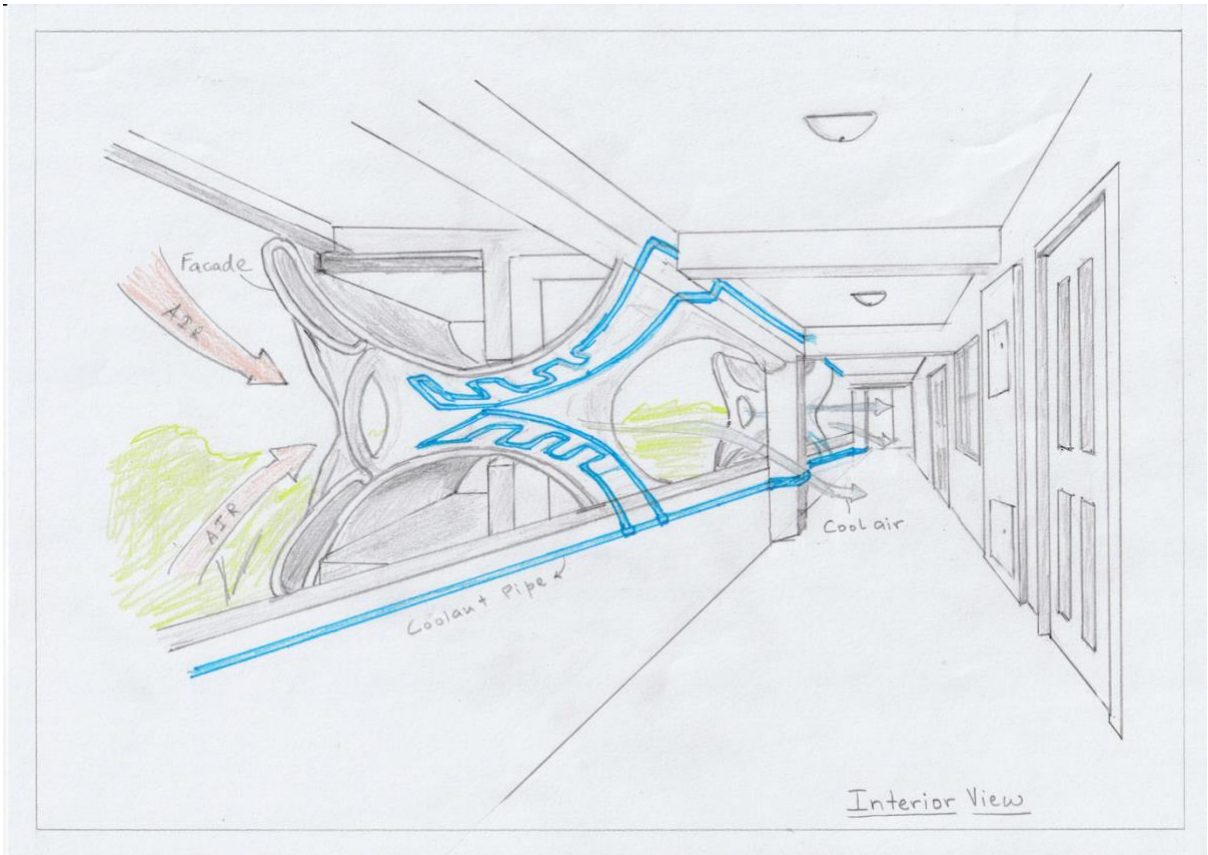
The idea was simple: we implement a design that is organic, yet acts as a modern looking facade on buildings which improve the thermal efficiency whilst keeping in mind the sustainability of this project.



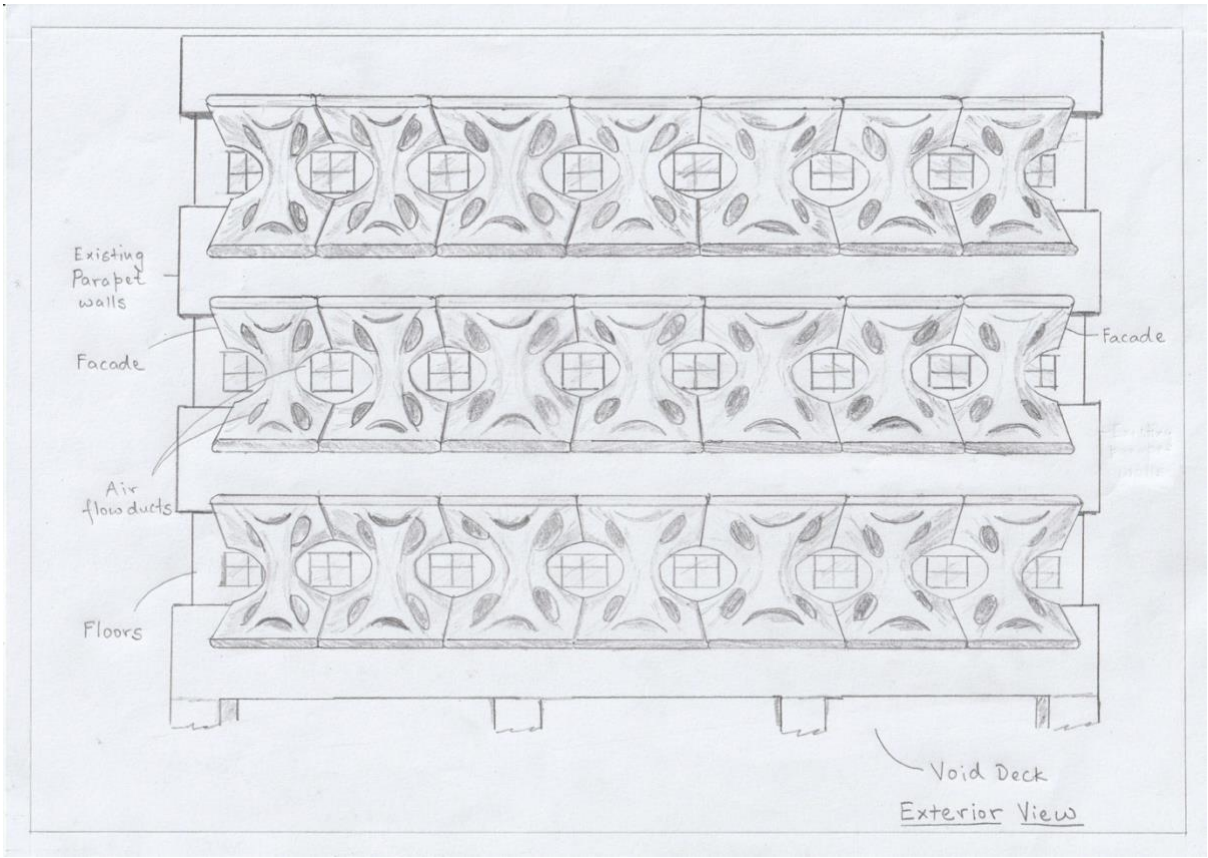
This rough idea was drafted using **Bernoulli's principle**. I wanted to see if I could take a fundamental law of fluid dynamics where air forced through a narrow constriction accelerates and drops in pressure. According to Bernoulli's principle, this sudden increase in speed creates a zone of low pressure within that narrow throat. This low-pressure zone acts as a natural vacuum. It continuously pulls more hot, humid air from the outside, creating a fan effect.

Using this we can manipulate the wind speed within one structure which was the core concept of this project.

The final design and its features



The final design manifests as a series of modular, horizontal "air scoops" that seamlessly integrate into the existing ventilation gaps between floors. By orienting the facade horizontally, we preserve the resident’s outward view while transforming the entire building exterior into a cooling façade. As strong, hot, and humid winds hit the exterior, the air is caught by the mouths of the facade and forced through the narrower coolant channels. Applying Bernoulli's principle, this constriction drastically accelerates the airflow and drops the pressure. This acceleration, known as the Venturi Effect, essentially acts as a vacuum, continuously pulling fresh air into the building without relying on mechanical fans to create a natural cooling breeze.



Construction

Material Innovation: 3D-Printed hydroceramics.

Creating the complex, aerodynamic Bernoulli geometries required for the facade would be incredibly wasteful and time consuming using traditional concrete casting. Instead, the design turns to additive manufacturing.

- **Large-Scale 3D Printing**: Inspired by construction technology innovators like Tvasta, who pioneer large-scale 3D printing for structural elements, the Aero-Weave modules are designed to be entirely 3D printed. This approach allows for the rapid, precise fabrication of the flowing Venturi funnels in a fraction of the time.
- **The Terracotta-Polymer Blend**: The material itself is an active participant in the cooling process. The printer uses a specialized terracotta-polymer blend, heavily influenced by the development of hydroceramics - an inexpensive, passive cooling material that successfully combines traditional clay with highly water-absorbent hydrogels.
- **Embedding coolant channel**: The coolant pipes are embedded during the 3D construction phase while keeping it in contact with the air outside for effective cooling.

Non-Destructive Installation The final, crucial aspect of this practical architecture is how it interfaces with the building. The Aero-Weave system is conceived as a lightweight, modular secondary skin. Because the air scoops are designed horizontally, they are fabricated to slot precisely into the existing horizontal ventilation gaps between the floor slabs of the building. This means the prefabricated panels can be hoisted by crane and seamlessly attached to the exterior facade without needing to alter, drill heavily into, or destroy the original standing structure. It is a rapid, scalable, and completely non-invasive urban retrofit.

Conclusion and Future Enhancements

The "Aero-Weave" facade system presents a sustainable, physics-based solution to the pervasive Urban Heat Island (UHI) effect in dense metropolises. By integrating principles of biomimicry and fluid dynamics, the project successfully moves beyond merely blocking sunlight to actively and passively cooling a structure.

Key Achievements of the "Aero-Weave" Design:

1. **Passive Cooling:** Achieves significant cooling and reduced reliance on electrical air conditioning, directly addressing **SDG 13 (Climate Action)** (see **Alignment with Sustainable Development Goals (SDGs)**).
2. **Sustainability:** The system incorporates a closed-loop water cycle, leveraging rainwater harvesting and greywater recycling, and avoids the use of toxic refrigerants (see **Usage of sustainable sources for coolants**).
3. **Practical Architecture:** The modular design, fabricated via large-scale 3D printing with a terracotta-polymer blend, is built for **non-destructive Installation**. It can be retrofitted onto existing buildings by slotting into the inter-floor gaps, making it a scalable, rapid urban upgrade (see **Modular Retrofitting (Practical Architecture)** and **Construction**).
4. **Scientific Validation:** The mechanism is grounded in fundamental laws of physics, making the cooling effect predictable and efficient (see **Conceptualisation**).

In conclusion, "Aero-Weave" is not just a building facade; it is a critical piece of infrastructure designed to increase the sustainability and thermal comfort of existing urban housing, aligning directly with **SDG 11 (Sustainable Cities and Communities)**.

Future Enhancements: Scaling and Optimization

To transition the "Aero-Weave" from a concept to a large-scale commercial reality, the following future enhancements should be prioritized:

1. Advanced Material Optimization

Further research is needed into the terracotta-polymer blend to maximize its performance as a passive evaporative cooling agent.

Enhancement Area	Objective
Hydrogel Concentration	Optimize the hydrogel and polymer ratio to increase cooling efficiency and reducing overall facade weight
Surface Texture	Introduce micro-scale surface textures (inspired by biomimetic research) to further increase the effective evaporative surface area within the Venturi channel.
Extending the time	Increasing the time it can go without fresh water supply.

2. Smart System Integration

While the core cooling is passive, smart integration can ensure optimal performance and resource management.

- **Sensor Integration:** Embed low-power integration with IOT to monitor metrics such as temperature, humidity, wind speed, and internal panel moisture levels.
- **Automated Louver Adjustment:** Develop a mechanism to allow residents or building managers to adjust the aperture of the air funnels based on wind direction and intensity, maximizing the Venturi effect for different conditions.
- **Water Management Control:** Implement smart valves to automatically divert excess rainwater and manage the flow of treated greywater into the closed-loop system, ensuring net zero water loss.

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